OLIST E-COMMERCE BUSINESS PERFORMANCE ANALYSIS

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Data Source: Olist Public E-Commerce Dataset (Kaggle)

1. EXECUTIVE SUMMARY

This report presents the key findings from our analysis of the Olist e-commerce business data. To conduct this analysis, we collected and organized data into a structured database, allowing us to accurately evaluate sales performance, customer behavior, product performance, and operational efficiency. During this process, data quality issues were identified and addressed to ensure reliable insights.

Our analysis highlights several important structural business findings:

- Revenue is highly concentrated among a relatively small number of product categories, following standard 80/20 (Pareto) patterns.
- The company maintains a healthy buffer between estimated and actual delivery times, helping ensure orders are delivered on time.
- However, delivery estimates may be more conservative than necessary, which increases shipping expectations shown to customers during checkout and directly contributes to cart abandonment.
- The findings identify major opportunities to improve customer experience, optimize delivery promises, and further scale gross platform velocity.

2. FINANCIAL PERFORMANCE BASELINE

To ensure accurate financial reporting and strict accounting integrity, cancelled and unavailable orders were programmatically filtered out from the analysis dataset. The platform's core performance baselines are established as:

GROSS REVENUE	COMPLETED ORDERS	AVERAGE ORDER VALUE
$13.49M	**98,199**	**$137.42**

REVENUE CONCENTRATION ANALYSIS (80/20 PRINCIPLE)

Although the marketplace offers items across **74 unique product categories**, systemic financial metrics indicate extreme volume density. Advanced cumulative calculations isolated the exact breakdown of this catalog structure:

- **18 categories (24% of the catalog)** generate **81.26% of total revenue**.
- These top departments account for roughly **$10.96 Million** in aggregate sales values.
- The remaining **56 long-tail categories** generate less than 19% of platform revenue despite consuming the majority of catalog space.

CATALOG REVENUE CONCENTRATION MATRIX

Below is the structured ranking view displaying the performance cliff and the precise Pareto cutoff point at category index position 18:

RANK	PRODUCT CATEGORY NAME (RAW / TRANSLATED)	GROSS REVENUE ($)	CUMULATIVE SHARE (%)
1	beleza_saude (Health & Beauty)	1,255,695.13	9.31%
2	relogios_presentes (Watches & Gifts)	1,198,185.21	18.18%
3	cama_mesa_banho (Bed, Bath & Table)	1,035,964.06	25.86%
4	esporte_lazer (Sports & Leisure)	979,740.92	33.12%
...	...	...	...
17	pcs (Computers / Systems)	222,963.13	79.68%
18	**pet_shop (Pet Supplies)** PARETO CUTOFF	**213,766.63**	**81.26%**
-	Remaining 56 Categories Combined	2,530,111.90	18.74%

Management Takeaway: Just 24% of product categories generate more than 81% of total revenue. Protecting and growing these key categories should be a top business priority, while lower-performing categories should be reviewed for layout optimization or selective consolidation.

3. DELIVERY PERFORMANCE & CUSTOMER SATISFACTION

To evaluate shipping efficiency, transit times, and customer sentiment metrics, we analyzed **96,470 successfully delivered orders**, cross-referencing actual transit lengths against customer checkout timelines.

ACTUAL TRANSIT SPEED	PROMISED WINDOW	EARLY ARRIVAL CUSHION
12.6 Days	**23.7 Days**	**11.2 Days**

The logistics infrastructure performs solidly, delivering orders an average of **11.2 days ahead of schedule**. However, displaying a delivery promise of nearly 24 days creates a major user-experience barrier at checkout, rendering the platform artificially slow compared to immediate market competitors.

TRANSIT VELOCITY VS. CUSTOMER SENTIMENT BREAKDOWN

Cross-referencing logistics logs with user reviews reveals a strict, non-linear psychological threshold regarding transit times:

REVIEW SCORE	VOLUME (ORDERS)	AVG. TRANSIT TIME	DAYS EARLY	USER SENTIMENT
⭐⭐⭐⭐⭐	56,810	10.7 Days	+12.7 Days	EXCELLENT
⭐⭐⭐⭐	18,943	12.3 Days	+11.7 Days	VERY GOOD
⭐⭐⭐	7,942	14.3 Days	+10.1 Days	NEUTRAL
⭐⭐	2,938	16.7 Days	+7.9 Days	DISSATISFIED
⭐	9,380	21.3 Days	+3.4 Days	CRITICAL DAMAGE

Customer Experience Insight: Customer tolerance drops off a cliff past the **14-day mark** in transit. When a delivery stretches out to an average of 21.3 days, a 1-star review is highly likely, even though the order technically arrived within the original promised checkout window.

4. DATA QUALITY & GOVERNANCE LOGS

Ensuring absolute pipeline and reporting integrity was a critical component of Phase 1. Two primary metadata transformations were injected into the SQL architecture to correct ingestion anomalies:

1. MISSING DELIVERY CONFIRMATION DATES

We isolated **8 distinct records** marked as officially "delivered" that completely lacked customer receipt timestamps. Temporal clustering analysis showed that 75% of these logs occurred between June and July 2018, confirming a systemic API database synchronization failure rather than standard manual data entry omissions. These rows were programmatically bypassed in transit-length arrays to guarantee accurate operations metrics.

2. UNCATEGORIZED PRODUCT ORPHAN RECOVERY

Our pipeline flagged **610 unique product IDs** that entered transactions completely missing an assigned product category. Left untreated, these sales metrics would drop out of index rankings. By applying a robust `COALESCE` fallback query string, these items were dynamically reclassified as `'Unassigned'`.

Data Governance Impact: The programmatic category recovery rule saved visibility for exactly **$178,572.55** in gross marketplace revenue, preserving metric tracking accuracy across corporate indices.

5. COMPLETE STRATEGIC RECOMMENDATIONS FRAMEWORK

OPERATIONAL WAREHOUSE REALIGNMENT

Based on the Pareto results, the top 18 categories must be prioritized within physical fulfillment centers. Relocating these high-velocity items to premium storage zones closest to processing stations will directly cut picker travel durations, lower packaging overhead, and speed up shipment initialization times.

CHECKOUT CONVERSION OPTIMIZATION

Compress the outward checkout shipping algorithm estimate from its current baseline of 23.7 days down to a dynamic 14 to 15-day range. This directly recaptures abandoned shopping carts by matching market speeds while safely preserving an operational buffer above our network's actual 12.6-day historical delivery average.

PROACTIVE CUSTOMER SERVICE TRIGGER ROUTING

Deploy an automated database trigger query that flags any order exceeding 14 days in active transit. This allows the customer support desk to implement proactive CRM recovery actions, offering tracking visibility and delay compensation vouchers before an unsatisfied user leaves a 1-star review on the platform.

6. FUTURE GROWTH OPPORTUNITIES (PHASE 2 ROADMAP)

To avoid scope creep and protect data integrity boundaries, specific database entities were left untouched during the Phase 1 audit. These datasets are scheduled for integration in the upcoming Phase 2 platform deployment:

TARGET ENTITY	UNDERLYING ATTRIBUTES	STRATEGIC APPLICATION
olist_order_payments	Installments, type metrics, voucher totals.	**Affordability Mechanics:** Correlate installment financing usage with higher Average Order Values to test Buy-Now-Pay-Later features.
olist_sellers	Merchant geographic states, city data points.	**Supply Bottleneck Isolation:** Determine whether transit delays stem from regional seller dispatch lag or carrier infrastructure deficits.
olist_customers & olist_geolocation	Lat/Long coordinates, regional postal codes.	**Spatial Hub Placement:** Model geospatial cluster densities to identify optimum regions for launching physical fulfillment hubs.